

Outline

1. Chapter 16: Other Acceptance Sampling Techniques

Reference: chapter 16 of the book

Acceptance Sampling by Variables

Advantages

- Same OC curve obtained with a smaller sample size
- Provides more information (data)
- When AQL is very small, samples sizes required by attribute sampling plans are very large

Disadvantages

- Distribution of OC curve must be known
- Separate sampling plan needed for each quality characteristic that is inspected
- A sample can be acceptable, but the lot rejected

Some Types of Variable Sampling Plans

1. **k-method:** Plans that control a lot or process parameter (usually the mean)
2. **M-method:** Plans that control the lot or process fraction defective (nonconforming)
3. **Continuous sampling:** Begin with 100% inspection and when a stated number of units are found to be free of defects, sampling inspection is instituted
4. **Skip-Lot sampling:** Provision made for inspecting only some fraction of lots submitted for inspection

MIL-STD-414 (ANSI/ASQC Z1.9)

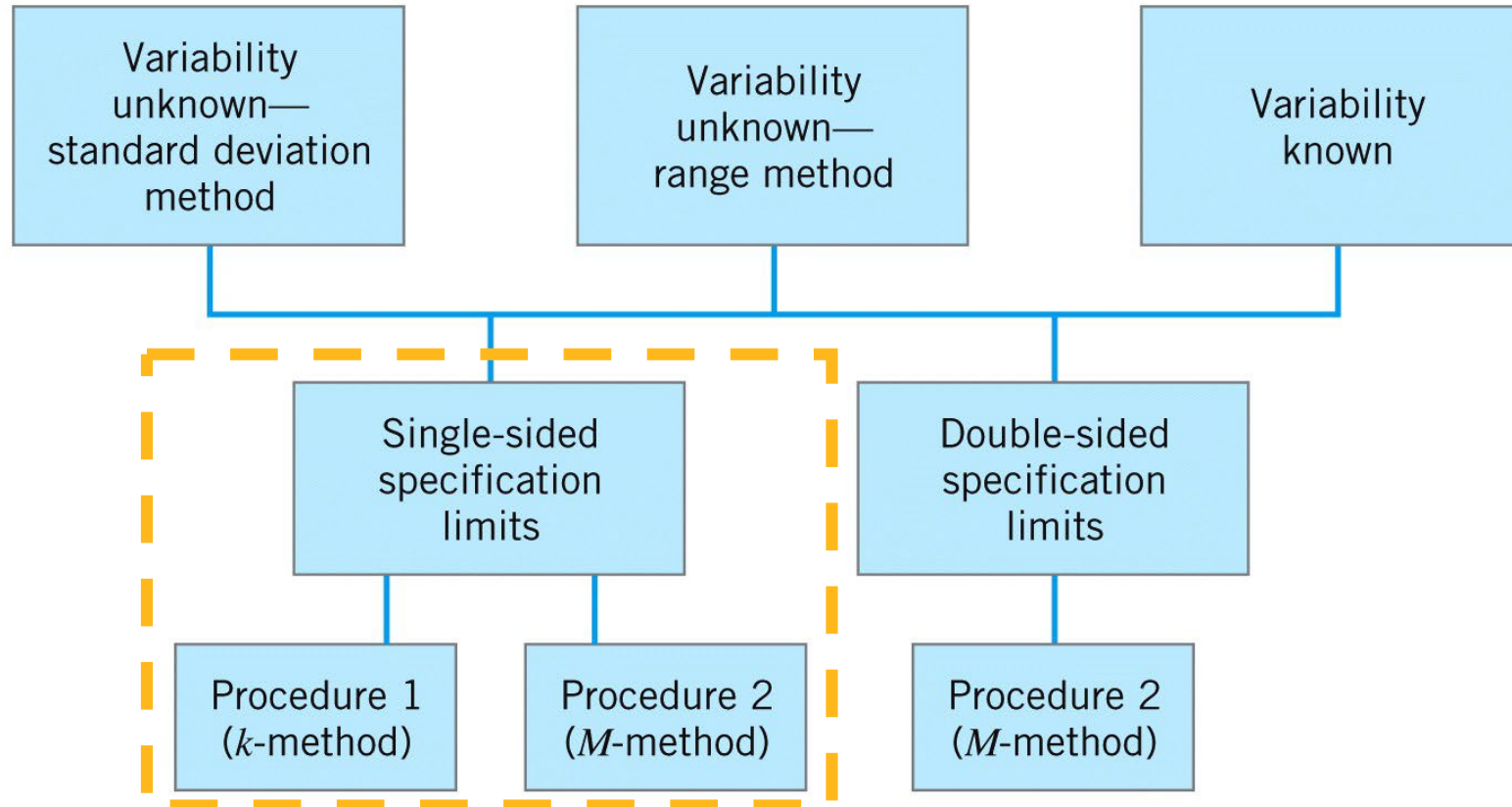


Figure 16.5
Courtesy of Doug Montgomery. © John Wiley & Sons, Inc. All rights reserved.

k-method

Procedure

1. Determine sample size code letter from Table A.2
2. Obtain values of sample size n and acceptability constant k from either Table B.1 or Table B.2
3. Compute $Z_{LSL} = \frac{\bar{X} - LSL}{\sigma}$
 - $Z_{LSL} \geq k \rightarrow$ accept the lot
 - Sample data imply lot is sufficiently above LSL
 - $Z_{LSL} < k \rightarrow$ reject the lot
 - Sample data imply lot is not sufficiently above LSL

k-method (2)

Level I
Level II
Level III
Level IV
Level V

Lower cost/greater risk

Greater cost/lower risk

■ TABLE 16.1

Sample-Size Code Letters (MIL STD 414, Table A.2)

Lot Size	Inspection Levels				
	I	II	III	IV	V
3 to 8	B	B	B	B	C
9 to 15	B	B	B	B	D
16 to 25	B	B	B	C	E
26 to 40	B	B	B	D	F
41 to 65	B	B	C	E	G
66 to 110	B	B	D	F	H
111 to 180	B	C	E	G	I
181 to 300	B	D	F	H	J
301 to 500	C	E	G	I	K
501 to 800	D	F	H	J	L
801 to 1,300	E	G	I	K	L
1,301 to 3,200	F	H	J	L	M
3,201 to 8,000	G	I	L	M	N
8,001 to 22,000	H	J	M	N	O
22,001 to 110,000	I	K	N	O	P
110,001 to 550,000	I	K	O	P	Q
550,001 and over	I	K	P	Q	Q

Table 16.1

Courtesy of Doug Montgomery. © John Wiley & Sons, Inc. All rights reserved.

k-method (3)

■ TABLE 16.2

Master Table for Normal and Tightened Inspection for Plans Based on Variability Unknown (Standard Deviation Method) (Single-Specification Limit—Form 1)(MIL STD 414, Table B.1)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (normal inspection)													
		.04	.065	.10	.15	.25	.40	.65	1.00	1.50	2.50	4.00	6.50	10.00	15.00
		<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>
B	3	↓	↓	↓	↓	↓	↓	↓	▼	▼	1.12	.958	.765	.566	.341
C	4	↓	↓	↓	↓	↓	↓	↓	1.45	1.34	1.17	1.01	.814	.617	.393
D	5	↓	↓	↓	↓	↓	↓	1.65	1.53	1.40	1.24	1.07	.874	.675	.455
E	7	↓	↓	↓	↓	2.00	1.88	1.75	1.62	1.50	1.33	1.15	.955	.755	.536
F	10	↓	↓	↓	2.24	2.11	1.98	1.84	1.72	1.58	1.41	1.23	1.03	.828	.611
G	15	2.64	2.53	2.42	2.32	2.20	2.06	1.91	1.79	1.65	1.47	1.30	1.09	.886	.664
H	20	2.69	2.58	2.47	2.36	2.24	2.11	1.96	1.82	1.69	1.51	1.33	1.12	.917	.695
I	25	2.72	2.61	2.50	2.40	2.26	2.14	1.98	1.85	1.72	1.53	1.35	1.14	.936	.712
J	30	2.73	2.61	2.51	2.41	2.28	2.15	2.00	1.86	1.73	1.55	1.36	1.15	.946	.723
K	35	2.77	2.65	2.54	2.45	2.31	2.18	2.03	1.89	1.76	1.57	1.39	1.18	.969	.745
L	40	2.77	2.66	2.55	2.44	2.31	2.18	2.03	1.89	1.76	1.58	1.39	1.18	.971	.746
M	50	2.83	2.71	2.60	2.50	2.35	2.22	2.08	1.93	1.80	1.61	1.42	1.21	1.00	.774
N	75	2.90	2.77	2.66	2.55	2.41	2.27	2.12	1.98	1.84	1.65	1.46	1.24	1.03	.804
O	100	2.92	2.80	2.69	2.58	2.43	2.29	2.14	2.00	1.86	1.67	1.48	1.26	1.05	.819
P	150	2.96	2.84	2.73	2.61	2.47	2.33	2.18	2.03	1.89	1.70	1.51	1.29	1.07	.841
Q	200	2.97	2.85	2.73	2.62	2.47	2.33	2.18	2.04	1.89	1.70	1.51	1.29	1.07	.845
		.065	.10	.15	.25	.40	.65	1.00	1.50	2.50	4.00	6.50	10.00	15.00	
Acceptable Quality Levels (tightened inspection)															

All AQL values are in percent defective.

↓ Use first sampling plan below arrow—that is, both sample size as well as *k* value. When sample size equals or exceeds lot size, every item in the lot must be inspected.

Table 16.2

Courtesy of Doug Montgomery. © John Wiley & Sons, Inc. All rights reserved.